Software Construction & Development

Semester Project Guidelines

Total Marks: 100

Semester Project Overview

Students are required to develop a complete software application using Java technologies learned during the course. The purpose of this project is to evaluate students' understanding of software development concepts including frontend design, backend logic, database connectivity, and software engineering practices.

Students will work:

- Individually

Allowed Technologies

Students must primarily use Java-based technologies.

Frontend Options

- Java Swing
- JavaFX

Backend / Logic

- Core Java
- Object-Oriented Programming Concepts
- JDBC

IDEs

- IntelliJ IDEA
- Eclipse

Project Phases

The semester project will be completed in TWO phases.

PHASE 1: Project Proposal / Idea Submission

Marks: 20

Students must submit a project proposal document containing the following:

Required Contents

1. Project Title

Name of the software/application.

2. Student Info

Include:

- Student Name
- Roll Number

3. Problem Statement

Explain:

- What problem the software solves
- Why the system is needed

4. Project Description

Brief explanation of:

- System functionality
- Main modules/features

5. Proposed Technologies

Mention:

- Java Technology Used
- Database
- IDE
- Libraries/Tools

6. Proposed Functionalities

List all planned features.

Example:

- Login System
- Add/Delete/Update Records
- Search Functionality
- Report Generation

- Dashboard

7. Methods / Classes to be Used

Students should mention possible:

- Classes
- Methods
- OOP Concepts
- Packages

Example:

- User Class
- DatabaseConnection Class
- addStudent()
- updateRecord()
- deleteItem()

PHASE 2: Final Project Submission

Marks: 80

Students must submit the complete working software along with documentation and presentation.

Mandatory Project Requirements

Every project MUST include:

Feature	Required
Java GUI	Yes
Database Connectivity	Yes
CRUD Operations	Yes
Login System	Yes
Input Validation	Yes
Search Feature	Yes

Feature	Required
Proper Exception Handling	Yes
OOP Concepts	Yes

OOP Concepts That Must Be Used

Students are required to demonstrate:

- Classes & Objects
- Encapsulation
- Inheritance
- Polymorphism
- Abstraction
- Constructors
- Method Overloading/Overriding

Final Submission Requirements

Students must submit:

1. Complete Source Code

- Proper folder structure
- Well-commented code

2. Executable Project

- Running application
- Database included

3. Documentation Report

The report must include:

1. Cover Page

2. Introduction
3. Objectives
4. Problem Statement
5. Screenshots
6. Testing
7. Conclusion
8. Future Enhancements

4. Presentation Slides

Students will present:

- Project idea
- Features
- Working demo
- Challenges faced

Mandatory GitHub Submission

All students are required to maintain and submit their project using GitHub.

GitHub Requirements

Each student must:

- Create a GitHub repository for the project
- Push code regularly throughout development
- Maintain proper commit history
- Upload complete source code
- Include project documentation
- Submit the final GitHub repository link

Repository Structure

The repository should contain:

- Source Code
- Screenshots

- Documentation Report
- Presentation Slides
- README.md file

README.md Must Include

Students must write a proper README file containing:

1. Project Title
2. Student Info
3. Project Description
4. Features
5. Installation Steps
6. Screenshots
7. Future Enhancements

GitHub Commit Guidelines

Students are encouraged to:

- Make regular commits
- Use meaningful commit messages
- Avoid uploading unnecessary files
- Maintain clean project structure

Important Rules

- Projects without GitHub repository submission may lose marks.
- Copying repositories from other students is strictly prohibited.
- Private repositories must provide instructor access.
- All group members should contribute to the repository.

Evaluation of GitHub Repository

Criteria	Marks
Proper Repository Structure	2

Criteria	Marks
Commit History	3
README Documentation	3
Code Organization	2

Total: 10 Marks

Evaluation Criteria

Phase 1: Proposal (20 Marks)

Criteria	Marks
Project Idea	5
Documentation Quality	5
Feasibility & Planning	5
Proposed Features	5

Phase 2: Final Submission (80 Marks)

Criteria	Marks
GUI Design	10
Java Logic & Functionality	15
Database Integration	15
OOP Concepts	10
CRUD Operations	10
Code Quality	5
Documentation	5
Presentation & Viva	10

Important Rules

- Copying projects from internet sources is strictly prohibited.
- Students must understand their complete code.
- Projects with no database connectivity will lose marks.
- Hardcoded-only projects are not acceptable.
- Each student must participate during viva.
- Late submissions may receive penalties.

Viva Questions May Include

Students should be able to explain:

- Database connectivity
- OOP concepts used
- GUI handling
- Event handling
- Exception handling
- Project architecture

Final Note

This project is intended to simulate real-world software development using Java technologies. Students are encouraged to focus on:

- Clean coding practices
- Proper software design
- Problem solving
- Professional presentation
- Creativity and innovation

Best of Luck!
